A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-26136466})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 1835 & 997 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,1835 + \mathbf{Z}\,(997 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 26136466 = 0$, of discriminant -104545864. Class group invariants $(60 \quad 10 \quad 5)$, class number 3000; the three stored generator ideals (HNF) are $\begin{pmatrix} 7 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5110 & 288 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + (\frac{209}{20805604708146875})\, s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2346748210944 = 2^8 \cdot 3^2 \cdot 163 \cdot 6248797$:

$$\begin{pmatrix}
12222646932 & 11519520870 & 4737325894 & 6286921160 & 5190493868 & 10955947053 & 10427831658 & 9817023442 & 3677316565 & 5326909892 \\
0 & 6 & 2 & 0 & 0 & 2 & 4 & 4 & 2 & 4 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 228053440845485870640; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9461046529216 = 2^6 \cdot 53 \cdot 103 \cdot 139 \cdot 194819$:

$$\begin{pmatrix}
591315408076 & 265660819632 & 521711836282 & 42802742505 & 151118098892 & 559501203342 & 417390893879 & 148028157570 & 520310520659 & 78272553440 \\
0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 708586214565857929334; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 6093y^3 - 28sy^2 + 840932y + 2672s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 12186*y^8 + 38806513*y^6 + 10243391992*y^4 - 3203685051888*y^
2 + 186603494470144
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\, s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $60665404266752 = 2^8 \cdot 53 \cdot 4471211989$:

$$\begin{pmatrix}
947896941668 & 803744554748 & 581557369948 & 946718864699 & 678689935669 & 380723959298 & 367805455490 & 675716128712 & 332960481410 & 700551049686 \\
0 & 4 & 0 & 2 & 3 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 403 factors with signed exponents of absolute value up to 828610105250069514479; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + (\frac{209}{20805604708146875})\, s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55886436734328 = 2^3 \cdot 3^2 \cdot 619 \cdot 1253958821$:

$$\begin{pmatrix} 4657203061194 & 4639511317372 & 2374126973256 & 3467527206912 & 2716291005782 & 971578147824 & 1257640549771 & 509573532860 & 1467219880982 & 3955367584420 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 & 2 & 2 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 330 factors with signed exponents of absolute value up to 88093979730; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $90033655327182 = 2 \cdot 3^2 \cdot 7 \cdot 714552820057$:

$$\begin{pmatrix} 30011218442394 & 896715638588 & 7520965343796 & 12442824857520 & 26563304891072 & 24956227602771 & 5431377399388 & 27709369433721 & 12473296436582 & 104536855666 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 333 factors with signed exponents of absolute value up to 86314277059; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s + 435)\, y^3 + (27s - 179284)\, y^2 + (202s + 3993818)\, y + (-6930s - 12439896)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 874*y^8 - 360308*y^7 + 35030463*y^6 - 1608201308*y^5
+ 44161485350*y^4 - 795530052256*y^3 + 11696814898196*y^2 - 17254014
7715376*y + 1409952178494216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1421394542708 = 2^2 \cdot 7^2 \cdot 7252012973$:

$$\begin{pmatrix} 101528181622 & 18304199278 & 75458192040 & 45029116622 & 70198358624 & 17452173720 & 67558878107 & 91857305288 & 16430314518 & 20342320430 \\ 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 7 & 1 & 5 & 6 & 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 117207437084; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\, y^2 + (48s - 2960434)\, y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10616973765804 = 2^2 \cdot 3^2 \cdot 7^2 \cdot 67 \cdot 2273 \cdot 39521$:

$$\begin{pmatrix}
1769495627634 & 1329768656350 & 158446249034 & 1175013398748 & 999515143070 & 965840048054 & 1217710622649 & 1602131266300 & 1279525768812 & 1036935003991 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 6 & 4 & 4 & 2 & 3 & 3 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 293 factors with signed exponents of absolute value up to 4970391044; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\, y^2 + (48s - 2960434)\, y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34336544468324 = 2^2 \cdot 7^2 \cdot 175186451369$:

$$\begin{pmatrix}
2452610319166 & 1780039898434 & 1388457492624 & 2286710704356 & 535664907641 & 1965598013425 & 1301819504696 & 486827927328 & 1478472405806 & 2045410874175 \\
0 & 14 & 7 & 0 & 9 & 0 & 2 & 0 & 6 & 8 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 111925686846; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 17y^3 + (-24s - 15189)\,y^2 + (48s - 2960434)\,y + (17616s + 62203704)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 33*y^8 - 30344*y^7 - 5890201*y^6 + 130844702*y^5 + 15
261557485*y^4 + 27598720452*y^3 - 15165395516780*y^2 - 3240996019296
96*y + 11980059248066112
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10616973765804 = 2^2 \cdot 3^2 \cdot 7^2 \cdot 67 \cdot 2273 \cdot 39521$:

$$\begin{pmatrix}
1769495627634 & 1329768656350 & 158446249034 & 1175013398748 & 999515143070 & 965840048054 & 1217710622649 & 1602131266300 & 1279525768812 & 1036935003991 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 6 & 4 & 4 & 2 & 3 & 3 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 290 factors with signed exponents of absolute value up to 65467097980; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\,y^3 + (-40s + 73531)\,y^2 + (151s + 1127436)\,y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \Big(\frac{209}{20805604708146875}\Big)\, s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $81512130868624 = 2^4 \cdot 2087 \cdot 23057 \cdot 105871$:

$$\begin{pmatrix}
20378032717156 & 696080944086 & 2756324354528 & 13880843975470 & 4947760541549 & 11777967861302 & 19951557836080 & 12262363762961 & 10818127748969 & 2137840979361 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 474 factors with signed exponents of absolute value up to 1598970961162; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\, y^3 + (-40s + 73531)\, y^2 + (151s + 1127436)\, y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $21470527694016 = 2^6 \cdot 3^4 \cdot 4141691299$:

$$\begin{pmatrix}
49700295588 & 11341174740 & 33197422236 & 32148654198 & 4740276406 & 43695304878 & 4763190692 & 32110319916 & 18464477219 & 35458116294 \\
0 & 6 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 6 & 0 & 4 & 2 & 5 & 1 & 2 & 5 \\
0 & 0 & 0 & 6 & 2 & 0 & 0 & 3 & 3 & 3 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 459 factors with signed exponents of absolute value up to 856926091686; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 1381)\, y^3 + (-40s + 73531)\, y^2 + (151s + 1127436)\, y + (11602s - 44474512)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 2753*y^8 + 155348*y^7 + 29857313*y^6 + 1792111018*y^5
+ 36484809669*y^4 - 633557471168*y^3 - 28932283470942*y^2 - 8707277
705800*y + 5496117919166408
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\, s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $372723413381824 = 2^6 \cdot 1609 \cdot 15359 \cdot 235661$:

$$\begin{pmatrix}
11647606668182 & 700271470046 & 9701728471874 & 5039156873040 & 559522516898 & 6574857273945 & 5572621467726 & 843398288634 & 4179136598726 & 3914810527097 \\
0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 460 factors with signed exponents of absolute value up to 1104732399212; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \left(\frac{209}{20805604708146875}\right) s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $163504363514560 = 2^6 \cdot 5 \cdot 71 \cdot 7196494873$:

$$\begin{pmatrix} 10219022719660 & 5226585942720 & 8298689620642 & 3049750371276 & 3025955700618 & 6337798042082 & 1673122811974 & 6318472220382 & 9367974016070 & 7453893073132 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 271 factors with signed exponents of absolute value up to 2268879515; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3778486223035968 = 2^6 \cdot 3^2 \cdot 17^2 \cdot 23^2 \cdot 71 \cdot 604343$:

$$\begin{pmatrix} 201325992276 & 48437162892 & 186550725962 & 173437118890 & 46967513682 & 187115361690 & 110212456565 & 131371418730 & 156492496497 & 2727625776 \\ 0 & 2346 & 920 & 1470 & 1656 & 2300 & 1478 & 1968 & 1096 & 1651 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 257 factors with signed exponents of absolute value up to 739988415; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 811y^3 - 10sy^2 + 2139000y + 18000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1622*y^8 + 4935721*y^6 - 855811400*y^4 - 4833806760000*y^2 +
8468214984000000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + \left(\frac{793}{2151402584438792}\right) s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $132544212316560 = 2^4 \cdot 3^2 \cdot 5 \cdot 11^2 \cdot 113 \cdot 13463701$:

$$\begin{pmatrix}
502061410290 & 104507961162 & 164171693970 & 151846119472 & 411528651695 & 80089077375 & 226205633178 & 21702133564 & 310306319760 & 73779729730 \\
0 & 22 & 10 & 4 & 6 & 1 & 16 & 20 & 18 & 11 \\
0 & 0 & 6 & 0 & 2 & 5 & 0 & 4 & 2 & 3 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 257 factors with signed exponents of absolute value up to 495252930; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b+c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s-1207)\,y^3 + (19s-62198)\,y^2 + (80s-499216)\,y + (2696s+2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{144237523}{20805604708146875} + \Big(\frac{209}{20805604708146875}\Big)\,s, \qquad J = \begin{pmatrix} 1835 & 997 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1835$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $151988111174592 = 2^6 \cdot 3^2 \cdot 7 \cdot 11^2 \cdot 29 \cdot 2539 \cdot 4231$:

$$\begin{pmatrix}
575712542328 & 404844853704 & 255684106696 & 456043302156 & 95109025090 & 27745198240 & 7446681596 & 357758666293 & 489047208417 & 5428937199 \\
0 & 6 & 4 & 3 & 0 & 3 & 2 & 0 & 5 & 3 \\
0 & 0 & 44 & 1 & 34 & 42 & 0 & 5 & 39 & 43 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 263 factors with signed exponents of absolute value up to 28494769746; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b+c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 1207)\, y^3 + (19s - 62198)\, y^2 + (80s - 499216)\, y + (2696s + 2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{12837823}{2638790623394114} + (-\frac{265}{10555162493576456})\, s, \qquad J = \begin{pmatrix} 2114 & 1252 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2114$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $40780756169216 = 2^9 \cdot 7^2 \cdot 37 \cdot 43932661$:

$$\begin{pmatrix}
45514236796 & 27366352076 & 37100911420 & 28887572238 & 13077434332 & 10886100648 & 44478105468 & 36457426951 & 43014106695 & 23094491949 \\
0 & 28 & 14 & 26 & 26 & 16 & 8 & 27 & 7 & 9 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 268 factors with signed exponents of absolute value up to 31362128195; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 1207)\, y^3 + (19s - 62198)\, y^2 + (80s - 499216)\, y + (2696s + 2528496)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2414*y^8 - 124396*y^7 + 26594883*y^6 + 1148388672*y^5 + 18690
797414*y^4 + 276379365504*y^3 + 2779583877408*y^2 + 8749694655488*y
+ 196363999679872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-2 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad -1 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{5708627}{537850646109698} + (\frac{793}{2151402584438792})\, s, \qquad J = \begin{pmatrix} 1538 & 1082 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1538$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $179749363531248 = 2^4 \cdot 3^2 \cdot 7 \cdot 859 \cdot 207593459$:

$$\begin{pmatrix}
7489556813802 & 584529739692 & 3385065733132 & 4187639608936 & 1334474812364 & 3397202460403 & 5083613412401 & 2425458178383 & 4961181185592 & 3341679638102 \\
0 & 6 & 0 & 2 & 1 & 5 & 1 & 1 & 4 & 2 \\
0 & 0 & 2 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 266 factors with signed exponents of absolute value up to 53441324936; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.